

① Applicazione, studio dei massimi e minimi locali

Fermat: se x_0 p.to di max/min locale interno $\Rightarrow f'(x_0)=0$ se f e' derivab

Cond. suff, cambio di segno di f'

$$f'(x_0) < 0 \quad x < x_0$$

$$f'(x_0) = 0 \quad \Rightarrow x_0 \text{ p.to minimo}$$

$$f'(x_0) > 0 \quad x > x_0$$



Ci serve il segno

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2}(x-x_0)^2 + R(x, x_0)$$

SE x_0 p.to di minimo locale $f(x) \geq f(x_0)$ x vicine a x_0
 $f(x) - f(x_0) \geq 0$ $x \in (x_0 - \delta, x_0 + \delta)$

Voglio $f(x) - f(x_0) \geq 0$ usando informaz su f''

$$f(x) - f(x_0) = (x-x_0)^2 \left[\frac{f''(x_0)}{2} + \frac{R_2(x, x_0)}{(x-x_0)^2} \right] \stackrel{?}{\geq} 0$$

$$(x-x_0)^2 \geq 0 \quad (x-x_0)^2 > 0 \quad x \neq x_0$$

$$\left[\frac{f''(x_0)}{2} + R_2(x, x_0) \right] \text{ Richiediamo } f''(x_0) \geq 0$$

Ci serve che tutta la [...] sia ≥ 0 .

(Teo p. segno $\lim_{x \rightarrow x_0} f''(x) = L > 0 \Rightarrow f''(x) > 0 \quad (x_0 - \delta, x_0 + \delta) \setminus \{x_0\}$)

$$f''(x) = \frac{f''(x_0)}{2} + \frac{R_2(x, x_0)}{(x-x_0)^2} \xrightarrow{x \rightarrow x_0} \frac{f''(x_0)}{2} + 0 = L$$

Se $f''(x_0) > 0 \Rightarrow \frac{f''(x_0)}{2} > 0$ e teo p. segno

$$\frac{f''(x_0)}{2}$$

Q55 $f'(x_0)=0$ $f''(x_0)=0$ $\rightarrow$ Boh?

$$f(x)=x^3 \quad x \in (-1, 1) \quad \begin{aligned} f'(x) &= 3x^2 \\ f''(x) &= 6x \end{aligned} \quad f'(x)=0 \Leftrightarrow x=0 \quad (x_0=0) \\ f''(x_0)=f''(0)=0$$



$$f(x)=x^4 \quad x \in (-1, 1) \quad \begin{aligned} f'(x) &= 4x^3 \\ f''(x) &= 12x^2 \end{aligned} \quad f'(x)=0 \Leftrightarrow$$



$$f(x) = x e^{-x} \quad x \geq 0$$

$$f \geq 0$$

$$f(0)=0 \quad 0 \text{ pt. di minimo assoluto}$$

$$f(x) \neq 0 \quad x \neq 0$$

$$\lim_{x \rightarrow +\infty} = \lim_{x \rightarrow +\infty} \frac{x}{e^x} = 0$$

$$f'(x) = e^{-x} + x e^{-x} (-1) = e^{-x} (1-x)$$

$$f'(x)=0 \Leftrightarrow 1-x=0 \Leftrightarrow x=1 \quad x_0=1 \quad f'(x_0)=0$$

① modo $f'(x) < 0 \Leftrightarrow e^{-x} (1-x) < 0 \Leftrightarrow 1-x < 0 \Leftrightarrow x > 1$

$$\begin{array}{ccccccc} + & + & 0 & - & - & & \\ \hline & & 1 & & - & & \end{array}$$

② modo $f''(x) = -e^{-x} (1-x) + e^{-x} (-1) = e^{-x} (1-x+1) = -e^{-x} (2-x)$

$$f''(1) = -e^{-1} (2-1) < 0 \quad x_0=1 \text{ è pt di max loc}$$

Formula di Taylor di grado $n \in \mathbb{N}$ relativa al punto x_0 con resto di Peano

$$f \in C^n(a, b) \quad x_0 \in (a, b)$$

$$f^{(n)}(x) = \frac{d^n}{dx^n} f(x) = f^{(n)}(x) \quad \text{2 volte}$$

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 + \frac{f^{(3)}(x_0)}{3!}(x-x_0)^3 + R_3(x-x_0)$$

$$\lim_{x \rightarrow x_0} \frac{R_3(x-x_0)}{(x-x_0)^3} = 0$$

$$= \sum_{k=0}^3 \frac{f^{(k)}(x_0)}{k!} (x-x_0)^k + R_3(x-x_0)$$

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x-x_0)^k + R_n(x, x_0)$$

$$= P_n(x, x_0) + R_n(x, x_0)$$

P_n è il polinomio di TAYLOR di grado n , relativo a x_0

$$\lim_{x \rightarrow x_0} \frac{R_n(x, x_0)}{(x-x_0)^n} = 0$$

$$R_n(x-x_0) = o((x-x_0)^n)$$

$$\Delta \quad f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x-x_0)^k + R_n(x, x_0)$$

Dimenticato di valutare in x_0

Es. $f(x) = e^x \quad x_0 = 1 \quad n = 3$

$$e^x = 1 + e^1(x-1) + \frac{e^1}{2!}(x-1)^2 + \frac{e^1}{3!}(x-1)^3 + R_3(x, 1)$$

$P_n(x)$

$$\lim_{x \rightarrow 1} \frac{R_3(x, 1)}{(x-1)^3} = 0$$

$$R_3(x, x_0) = o((x-1)^3)$$

• $f(x) = \sin(x) \quad x_0 = 0 \quad n = 6$

$$\sin(x) = P_6(x, 0) + R_6(x, 0)$$

$$= f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5 + \frac{f^{(6)}(0)}{6!}x^6 + R_6(x, 0)$$

$$= 0 + x + 0 - \frac{x^3}{3!} + 0 + \frac{x^5}{5!} + 0 + R_6(x, 0)$$

$$f(x) = \sin x \quad \text{è dispari}$$

$$f'(x) \quad \text{è pari}$$

$$f''(x) \quad \text{è dispari}$$

$$\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + R_6(x, 0)$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + R_3(x, 0)$$

$$f(x) = \underbrace{f(x_0) + f'(x_0)(x-x_0)}_{\varphi(x)} + \frac{f''(x_0)}{2!}(x-x_0)^2 + \frac{f'''(x_0)}{3!}(x-x_0)^3 + R_3(x-x_0)$$



Def x_0 PT. DI FLESSO se $\varphi(x) = f(x_0) + f'(x_0)(x-x_0)$
 Retta tg al grafico in $(x_0, f(x_0))$

$$f(x) > \varphi(x) \quad x_0 - \delta < x < 0$$

$$f(x) < \varphi(x) \quad x_0 + \delta > x > 0$$

oppure

$$f(x) < \varphi(x) \quad x < x_0$$

$$f(x) > \varphi(x) \quad x > x_0$$

$$x < x_0$$

$$x > x_0$$

$$x \in (x_0 - \delta, x_0 + \delta)$$

$$f(x) - [f(x_0) + f'(x_0)(x-x_0)] =$$

TEOR. C.S. per flesso $f \in C^3(a, b)$ $x_0 \in (a, b)$

$$f''(x_0) = 0 \quad \text{e} \quad f'''(x_0) \neq 0 \quad \Rightarrow \quad x_0 \text{ è pto di flesso}$$

$$f(x) = x^4$$

$$f'(x) = 4x^3$$

$$f''(x) = 12x^2$$

$$f'''(x) = 24x$$

$$f''(x) = 0 \quad \Leftrightarrow \quad x = 0$$

$$f'''(0) = 0$$

minimo (no flesso)

$$f(x) = x^5$$

$$f''(x)$$

Ripasso Taylor Rabasa

Obiettivo: Usare Taylor per il calcolo dei limiti.

Ripasso
formula

Sia $f: (a,b) \rightarrow \mathbb{R}$ una f. derivab n-volte in $x_0 \in (a,b)$
e in un suo intorno

La formula di Taylor centrata in x_0 con resto di Peano e'

$$f(x) = P_n(x) + R_n(x)$$

dove

- il polinomio

$$P_n(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} \underbrace{(x-x_0)^k}_{\text{centrata in } x_0}$$

$$f_{(1)}^{(0)} = f(x)$$

- e il Resto di Peano e'

$$R_n(x) = o((x-x_0)^n) \rightarrow \text{quando } x \rightarrow x_0$$

Osservaz il resto $o((x-x_0)^n)$ si dice l'errore infinitesimo di ordine superiore a $(x-x_0)^n$

ESEMPIO Calcolare i primi termini del polin. in $x_0 = 0$

1) $f(x) = e^x$

poiché $e^{x^{(k)}} = e^x$

$$\rightarrow f_{(0)}^{(k)} = 1 \quad \forall k \in \mathbb{Z}$$

$$\rightarrow P_n(x) = \sum_{k=0}^n \frac{x^k}{k!}$$

$$P_n(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!}$$

$$2) f(x) = \sin(x)$$

$$f(0) = \sin(0) = 0$$

$$f'(0) = \cos(0) = +1$$

$$f''(0) = -\sin(0) = -0 = 0$$

$$f'''(0) = -\cos(0) = -1$$

$$f^{(4)}(0) = \sin(0) = 0$$

$$\rightarrow P_n(x) = \sum_{k=0}^n \frac{(-1)^k}{(2k+1)!} \cdot x^{2k+1}$$

$$P_n(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!}$$

$$3) f(x) = \ln(1+x) \quad x > -1$$

$$f'(x) = \frac{1}{1+x} \rightarrow f'(0) = 1$$

$$f''(x) = -\frac{1}{(1+x)^2} \rightarrow f''(0) = -1$$

$$f'''(x) = 2 \frac{1}{(1+x)^3} \rightarrow f'''(0) = 2$$

$$f^{(4)}(x) = -6 \frac{1}{(1+x)^4} \rightarrow f^{(4)}(0) = -6$$

$$f^{(n)}(x) = (-1)^{n-1} \frac{(n-1)!}{(1+x)^n}$$

$$P_n(x) = 0 + x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} \dots + (-1)^{n-1} \frac{x^n}{n}$$

$$P_n(x) = \sum_{k=0}^n \frac{x^k}{k} (-1)^{k-1}$$

$$4) f(x) = \frac{1}{1-x} \quad ; \quad |x| < 1$$

$$f(0) = 1$$

$$f'(x) = -1 \frac{1}{(1-x)^2} (-1)$$

$$= \frac{1}{(1-x)^2} \rightarrow f'(0) = 1$$

$$f''(x) = -2 \frac{1}{(1-x)^3} \rightarrow f''(0) = -2$$

$$f'''(x) = 1 \cdot 2 \cdot 3 \frac{1}{(1-x)^4} (-1) \rightarrow f'''(0) = -6$$

$$f^{(k)}(x) = k! \cdot \frac{1}{(1-x)^{k+1}} \rightarrow f^{(k)}_{(0)} = k!$$

$$P_n(x) = 1 + x + x^2 + x^3 + \dots + x^n$$

Allora $\left(\frac{1}{1-x}\right) - (1 + x + x^2 + x^3 + \dots + x^n) = o(x^n)$

anche $(1-x)(1 + x + x^2 + x^3 + \dots + x^n) =$
 $= 1 - x^{n+1}$

$$\rightarrow \frac{1}{1-x} - P_n(x) =$$

$$= \frac{1 - (1 - x^{n+1})}{1-x} = \frac{x^{n+1}}{1-x}$$

Applicazioni Quando ci troviamo con un limite indeterminato dove il det. è un polinom., la formula di Taylor Permette di sostituire la funz. (Numeratore) con i loro polinomi (di Taylor) equivalenti

$$\lim_{x \rightarrow 0} \frac{f(x)}{x^n} = \lim_{x \rightarrow 0} \frac{P_n^f(x)}{x^n}$$

$$\frac{f(x)}{x^n} = \frac{P_n^f(x) + o(x^n)}{x^n}$$

$$\Rightarrow$$

$$\lim_{x \rightarrow 0} \frac{f(x)}{x^n} \stackrel{!}{=} \lim_{x \rightarrow 0} \frac{P_n^f(x)}{x^n}$$

Es

$$1) \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = \lim_{x \rightarrow 0} \frac{1 + x - 1}{x} = \lim_{x \rightarrow 0} \frac{x}{x} = 1$$

Grado del denominatore $n=1$

$$\rightarrow P_1(x) = 1 + x$$

$$2) \lim_{x \rightarrow 0} \frac{\sin(x^2)}{x^2} \quad \left[\frac{0}{0} \right]$$

• Grado den $n=2$

$$\begin{aligned} \text{Sviluppo } P_2(x) &= x - \frac{x^3}{3!} + \frac{x^5}{5!} \\ &= x + o(x^2) \end{aligned}$$

$$\lim_{x \rightarrow 0} \frac{-\frac{x^3}{3!} + \frac{x^5}{5!}}{x^2} = \lim_{x \rightarrow 0} -\frac{x}{3!} + \frac{x^3}{5!} = 0$$

• $t = x^2$

$$\begin{aligned} \sin(t) &= P_2(t) + o(t^2) \\ &= t + \underbrace{o(t) + o(t^2)}_{= o(t)} \end{aligned}$$

$$\begin{aligned} \sin(x^2) &= P_2(x^2) + o(x^2) \\ &= \overset{\downarrow}{x^2} + \underbrace{o(x^2) + o(x^4)}_{\text{si considera ordine minore}} \\ &= x^2 + o(x^2) \end{aligned}$$

$$\begin{aligned} \Rightarrow \lim_{x \rightarrow 0} \frac{\sin(x^2)}{x^2} &= \lim_{x \rightarrow 0} \frac{x^2 + o(x^2)}{x^2} \\ &= 1 + \underbrace{\lim_{x \rightarrow 0} \frac{o(x^2)}{x^2}}_{=0} \quad \text{Per def di ordine} \end{aligned}$$

$$3) \lim_{x \rightarrow 0} \frac{\sin(x) - x}{x^3}$$

• Grado denom $n=3$

• $P_3(x) = x - \frac{x^3}{3!} + o(x^3)$

$$\lim_{x \rightarrow 0} \frac{\sin(x) - x}{x^3} =$$

$$= \lim_{x \rightarrow 0} \frac{P_3(x) + o(x^3) - x}{x^3} =$$

$$= \lim_{x \rightarrow 0} \left[\frac{P_3(x) - x}{x^3} + \frac{o(x^3)}{x^3} \right] = \overset{\substack{\text{se} \\ \text{esiste}}}{\lim_{x \rightarrow 0} \frac{x - \frac{x^3}{3!} - x}{x^3}} + \lim_{x \rightarrow 0} \frac{o(x^3)}{x^3} = 0$$

$$= -\frac{1}{3!} = -\frac{1}{6}$$

$$4) \lim_{x \rightarrow 0} \frac{\ln(1+x) - x \cos(x)}{x^2}$$

• grado den $n=2$

$$P_2^{\ln(1+x)} = x - \frac{x^2}{2}$$

$$P_1^{\cos x} = 1 + o(x)$$

possiamo scrivere il numeratore come

$$\ln(1+x) = x - \frac{x^2}{2} + o(x^2)$$

$$\cos(x) = 1 + o(x)$$

$$\ln(1+x) - x \cos(x) = x - \frac{x^2}{2} + o(x^2) + x(-1) + o(x)$$

$$= \cancel{x} - \frac{x^2}{2} + o(x^2) - \cancel{x} + \underbrace{x o(x)}_{o(x^2)}$$

$$\ln(1+x) - x \cos(x) = -\frac{x^2}{2} + o(x^2)$$



Determino Max \ Min locale

idea: usare deriv seconda

↳ il primo coeff non nullo del polin. di Tay

1) $f(x) = \sin(x^2) - x^2$

analisi del punto $x_0 = 0$

Determinare la natura del pto STAZIONARIO $x_0 = 0$

- Verifica della STAZIONARITÀ

$$f'(0) = \cos(x^2) \cdot 2x - 2x \big|_0$$

$$= (\cos(0) - 1) \cdot 2(0) = 0$$

- Sviluppo di Taylor

- $f(0) = 0$

- $f'(0) = 0$

- $f''(0) = (-\sin(x^2)(2x^2) + \cos(x^2)2 - 2) \big|_0$

$$= 0 + 2 - 2 = 0$$

- $f'''(x) =$

ORDINE Di Funzione

- Concetto di
- O grande
 - o piccolo

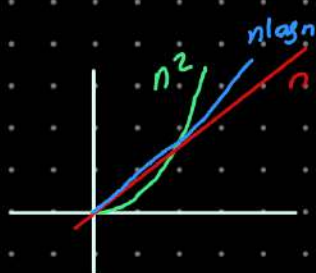
Ordinamento di una lista (più grande)

es: (Exel, data, cognome ecc)

- Algoritmo 1 (Bubble sort) $\sim n^2$
- Alg. 2. (Merge) $\sim n \log(n)$

Si dirà Alg. 1 = $O(\text{Alg. 2})$

↳ quando $n \rightarrow +\infty$
la lista è più grande



DEF. O grande

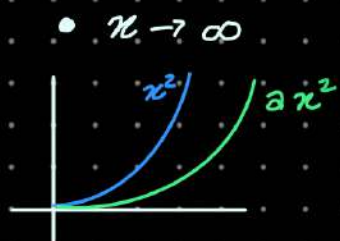
Notazione: $f(x) = O(g(x))$

indica un limite [fino a costante moltiplicativa]:
non nulla

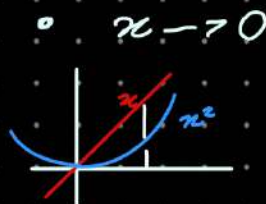
- quando $x \rightarrow \pm\infty$ non dipende dal segno

$$\lim_{x \rightarrow +\infty} \frac{|f(x)|}{|g(x)|} = L, \quad \begin{matrix} L \neq 0 \\ L > 0 \end{matrix}$$

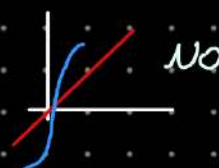
- Quando $x \rightarrow 0$ $\frac{|f(x)|}{|g(x)|} = L$ $L > 0$



il loro
rapp. è
sempre $\frac{1}{2}$



No



No

ES

① $f_1 = O(g)$
quando $x \rightarrow \infty$?
No $\rightarrow$ il $\frac{f(x)}{g(x)} \rightarrow \inf$

• $g(x) = x^2$

$f_1(x) = x^3 \Rightarrow$

② $f_1 = O(g)$
quando $x \rightarrow 0$?
No

• $x^n = O(x^m)$ quando $x \rightarrow 0$

se e solo se $n = m$

$a_1 x^{m_1} + \dots + a_r x^{m_r} = O(b_1 x^{n_1} + \dots + b_r x^{n_r}) \Leftrightarrow m_1 = n_1$

Domanda $f = O(g) \Rightarrow g = O(f)$? Si $\checkmark$
 $\hookrightarrow$ il rapp. non impare

Def.

o piccolo

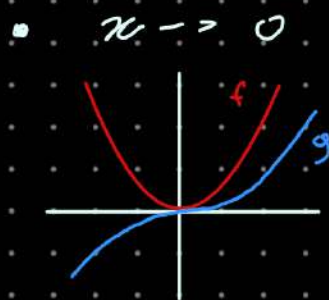
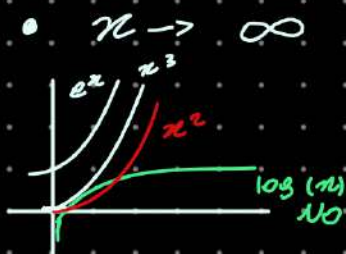
La notaz $f(x) = o(g(x))$ definisce

il limite STRETTO

• $x \rightarrow +\infty$
 $(x \rightarrow -\infty)$ $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = 0$

Significato Intuitivo la funz f decresce strettamente più lentamente di g

$\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = 0$



Proprietà

• $x^2 = O$



- Se $f(x) = O(g_1(x))$
 $h(x) = O(g_2(x))$

$$\Rightarrow f(x) + h(x) = O(\max(g_1, g_2)) \quad (\rightarrow +\infty)$$

$$\bullet O(x^2 + x^3) = O(x^3) \quad \rightarrow +\infty$$

$$\bullet O(x^2 + a^x) = O(a^x) \quad \rightarrow +\infty \quad (a > 0)$$

$$\bullet O(x^2 + a^x) = O(a^x) \quad \rightarrow 0$$

$$\bullet O(x^2 + \ln x) = O(a^x) \quad \rightarrow +\infty$$

$$\bullet O(x^2 + |\ln x|) = O(\ln x) \quad \rightarrow 0$$

PRODOTTI

$$\bullet O(g_1) \cdot O(g_2) = O(g_1 \cdot g_2)$$

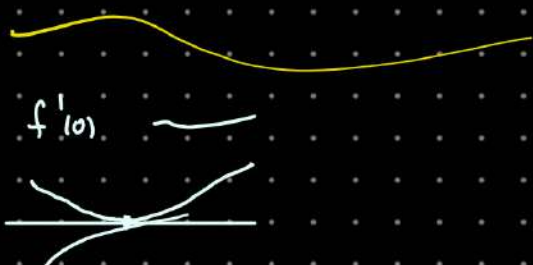
$$\bullet O(g_1) \cdot o(g_2) = o(g_1 \cdot g_2)$$

$$\rightarrow \bullet \underbrace{O(g_1)}_n \cdot \underbrace{o(g_2)}_{\text{zero}} = \underbrace{o}_{0}(g_1 \cdot g_2)$$

ES • $O(x) \cdot o(x) = o(x^2)$

$$O(1) \ll O(\ln n) \ll O(n^\epsilon) \ll O(n^c) \ll \dots$$

ES



$$\sin(t) = t - \frac{t^3}{6} + o(t^3)$$

$$= t + o(t)$$

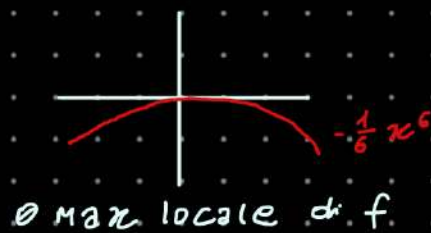
Poniamo $t = x^2$

$$\sin(x^2) = x^2 - \frac{x^6}{6} + o(x^6)$$

$$= x^2 + o(x^2)$$

$$f(x) = \sin(x^2) - x^2 = x^2 - \frac{x^6}{6} + o(x^6) - x^2$$

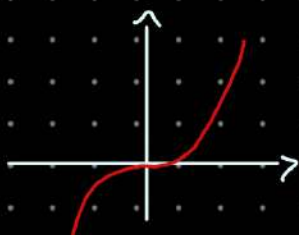
$$= -\frac{1}{6}x^6 + o(x^6)$$



$$\bullet f(x) = e^x - \left(1 + \frac{x^2}{2}\right)$$

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + o(x^3)$$

$$f(x) = \frac{x^3}{6} + o(x^3)$$



$x_0 = 0$ è p.to di flesso

14/11/25

$$f(x) = \sum_{k=0}^n \underbrace{\frac{f^{(k)}(x_0)}{k!} (x-x_0)^k}_{\text{P. Taylor}} + R(x, x_0)$$

$$+ R(x, x_0)$$

Resto di Peano

$$\lim_{x \rightarrow x_0} \frac{R_n(x, x_0)}{(x-x_0)^n} = 0$$

$$R_n(x, x_0) = o(x-x_0)^n$$

non da nessuna inf. sulla "taglia" di R_n .

$$R_n(x, x_0) = (x-x_0)^{n+1} = 10^{10^{10}} (x-x_0)^{n+2}$$

Teorema $f'(x_0) = 0$ $f''(x_0) > 0$ $\Rightarrow x_0$ p.to min. locale
 $x_0 \in (a, b)$

$$f(x) - f(x_0) = (x-x_0)^2 \left[\frac{f''(x)}{2} + \frac{R_2(x, x_0)}{(x-x_0)^2} \right]$$

$$\left\{ \begin{array}{l} f'(x_0) = 0 \\ f''(x_0) = 0 \\ f'''(x_0) = 0 \\ f^{(4)}(x_0) > 0 \end{array} \right. \Rightarrow f(x) = f(x_0) +$$

COROLLARIO $f'(x_0) = f''(x_0) = \dots = f^{(2n-1)}(x_0) = 0$ e

la prima derivata PARI $f^{(2n)}(x_0) > 0 \Rightarrow x_0$ p.t. min. loc.

non nulla $f^{(2n)}(x_0) < 0 \Rightarrow x_0$ p.t. Max. loc

LAGRANGE

$$\frac{f(x) - f(x_0)}{x - x_0} = f'(c)$$

$$|f(x) - f(x_0)| \leq |f'(c)| |x - x_0|$$

$$|f(x) - f(x_0)| = f'(c)(x - x_0)$$

$$f(x) = f(x_0) + f'(c)(x - x_0)$$

$$f(x) = \sum_{k=0}^n \underbrace{\frac{f^{(k)}(x_0)}{k!} (x - x_0)^k}_{P_n \text{ Taylor}} + \underbrace{\frac{f^{(n+1)}(c)}{(n+1)!} (x - x_0)^{n+1}}_{\text{Resto di LAGRANGE}} \quad 0 < |x_0 - c| < |x - x_0|$$

$f \in C^{n+1}(a, b)$ $x_0 \in (a, b)$ f derivabile $n+1$ volte

ES

$$\sin(1^\circ) = \sin\left(\frac{\pi}{180}\right) \quad \sin\left(\frac{\pi}{180}\right) \approx 0$$

$$n=1 \quad f(x) = f(0) + f'(0)(x-0) + \frac{f''(c)}{2!} (x-0)^2$$

$$\sin(x) = \sin 0 + \cos(0) \cdot x + \left| -\frac{\sin(c)}{2!} x^2 \right|$$

$$\sin\left(\frac{\pi}{180}\right) = 0 + \frac{\pi}{180} + \left| -\frac{\sin(c)}{2!} \left(\frac{\pi}{180}\right)^2 \right|$$

$$\left| \sin\left(\frac{\pi}{180}\right) - \frac{\pi}{180} \right| = \left| -\frac{\sin(c)}{2!} \left(\frac{\pi}{180}\right)^2 \right| = \frac{1}{2} \frac{\pi^2}{180^2} |\sin(c)| \leq \frac{1}{2} \left(\frac{\pi}{180}\right)^2$$

$$\leq 2 \left| \frac{4 \cdot 4}{4 \cdot 45} \right| = \frac{1}{2} \frac{1}{45^2}$$

$$180 = 10 \cdot 18 = 2 \cdot 5 \cdot 2 \cdot 9$$

$$n=2 \quad f(x) = f(0) + \underbrace{f'(0)x + \frac{f''(0)x^2}{2}}_{P_2} + \frac{f'''(c)}{3!} x^3$$

R_2 Lagrange

$$\sin(x) = \sin(0) + x + 0 + \left| -\frac{\cos(c)}{3!} x^3 \right|$$

$$n=3$$

$$n=4$$

ES $f(x) = \ln(x+1)$

ty in $x_0=0$

$$f'(x) = \frac{1}{1+x} = (1+x)^{-1}$$

$$f''(x) = -(1+x)^{-2}$$

$$f'''(x) = 2(1+x)^{-3}$$

$$f^{(4)}(x) = -3 \cdot 2 (1+x)^{-4}$$

$$f^{(5)}(x) = 4 \cdot 3 \cdot 2 (1+x)^{-5}$$

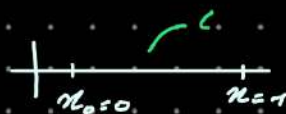
$$f^{(n)}(x) = \frac{d^n}{dx^n} \ln(x+1) = (-1)^{n-1} (n-1)! (1+x)^{-n}$$

(ty in $x_0=0$) $f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{4!}x^3 + f^{(4)} + R_5$

$$= 0 + x - \frac{x^2}{2} + x \frac{x^3}{3 \cdot 2 \cdot 1} - \frac{(3 \cdot 2)x^4}{4 \cdot 3 \cdot 2} + \frac{4 \cdot 3 \cdot 2 (1+x)^{-5}}{5 \cdot 4 \cdot 3 \cdot 2} x^5$$

$x=1$ $f(1) = \ln(1+1) = \ln 2 = 1 + \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} \frac{1}{(1+1)^5}$

$$\ln 2 - (1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4}) = \frac{1}{5} \frac{1}{(1+1)^5}$$



$$\ln 2 - (1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4}) \leq \frac{1}{5}$$

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x-x_0)^k + \frac{f^{(n+1)}(\xi)}{(n+1)!} (x-x_0)^{n+1}$$

$$f'(x) \geq 0 \quad \text{in } (a,b) \Rightarrow f \text{ crescente}$$

$$f'(x) > 0 \quad \text{in } (a,b) \Rightarrow f \text{ e' strett. cresc.}$$

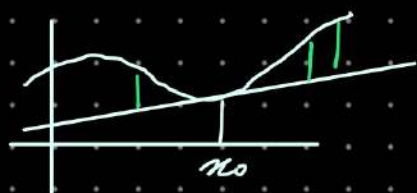
$$\frac{f(x) - f(x_0)}{x - x_0} = f'(\xi) \geq 0 \Rightarrow \text{Rapp. incr. e' } \geq 0$$

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + f''(\xi) \frac{(x-x_0)^2}{2}$$

$$f(x) - [f(x_0) + f'(x_0)(x-x_0)] = f''(\xi) \frac{(x-x_0)^2}{2}$$

r. tang. in x_0

$$f(x) - [f(x_0) + f'(x_0)(x-x_0)] \geq 0$$



$$f(x) \geq f(x_0) + f'(x_0)(x-x_0) \quad \forall x \in (a,b) \quad x_0 \in (a,b)$$

$$f \in C^2(a,b) \quad f''(x) \geq 0 \quad \forall x \in (a,b)$$

Osserv Se $f''(x) \geq 0 \quad \forall x \in (a,b)$

$$f(x) - [f(x_0) + f'(x_0)(x-x_0)] \geq 0 \quad \forall x \in (a,b) \quad \forall x_0 \in (a,b)$$

In (a,b) la funz. f "sta sopra" alla r. tang. in x_0 qualsiasi sia il pto $x_0 \in (a,b)$

DEF f e' derivabile, si dice **CONVEXA** in (a,b) se

$$f(x) \geq f(x_0) + f'(x_0)(x-x_0) \quad \forall$$

Teo. $f \in C^2(a,b)$ e $f''(x) \geq 0$ in $(a,b) \Rightarrow f$ e' **CONVEXA** (a,b)

≤ 0 CONCAVA

DEF f è concava in (a,b) se $f(x) \leq f(x_0) + f'(x_0)(x-x_0)$
 $\forall x, x_0 \in (a,b)$

$$f(x) = \ln|x| \quad x \neq 0$$

$$f'(x) = \frac{1}{x} \quad x \neq 0$$

$$f''(x) = -\frac{1}{x^2} < 0 \quad x \neq 0 \quad \Rightarrow f \text{ concava in } \mathbb{R} \setminus \{0\}$$

DEF f è strettamente convessa se $f(x) > f(x_0) + f'(x_0)(x-x_0)$
in (a,b)



$$f''(x) > 0 \quad \text{in } (a,b) \Rightarrow f \text{ è str. conv. in } (a,b)$$

Def x_0 è un p.to di flesso se $x < x_0$ $f(x) \leq f(x_0) + f'(x_0)(x-x_0)$
 $x > x_0$ $f(x) \geq f(x_0) + f'(x_0)(x-x_0)$

• $f \in C^2(a,b)$ 2 volte COND. NECES. per flesso in x_0 è $\sum$
ma non suff.

$$f'' \quad \text{Se } f''(x_0) > 0 \Rightarrow f''(x) > 0 \text{ in } (x_0 - \delta, x_0 + \delta) \\ \Rightarrow f \text{ conv. in } =$$

• COND. SUFFIC. per flesso in x_0 $f''(x_0) = 0$ $f'''(x_0) \neq 0$

$$f(x) = x e^{-x} \quad x \geq 0$$

$$f'(x) = e^{-x} + x e^{-x} (-1) = \\ = e^{-x} (1-x)$$

$$f''(x) = -e^{-x} (1-x) + e^{-x} (-1) \\ = e^{-x} (x-2)$$



$x_1 = 2$ è p.to flesso A tang
non orizz.

$$f'(2) = e^{-2} (1-2) = -\frac{1}{e^2}$$

se f è str. conv. in $\mathbb{R}$ allora interseca una retta in al più due punti



$$f(x) \quad g(x) = mx + q$$

PER ASSUNDO

Suppongo di avere 3 pt:

$$f(x) - g(x) = 0 \quad x = x_1, x_2, x_3$$

$$f(x_1) = g(x_1) \dots$$

$$f(x) - mx + q = 0 \quad x = x_1, x_2, x_3$$

Per Rolle in $[x_1, x_2] \Rightarrow x_1 < c_1 < x_2$

in $[x_2, x_3] \Rightarrow x_2 < c_2 < x_3$



$$F'(c_1) = 0 \quad F'(c_2) = 0$$

$F'(x)$ si annulla in c_1, c_2

Rolle A $F' \Rightarrow c_1 < c_2 < c_2$

$$(F')'(c) = 0$$

$$F''(c) = 0$$

$$F(x) = f(x)$$